

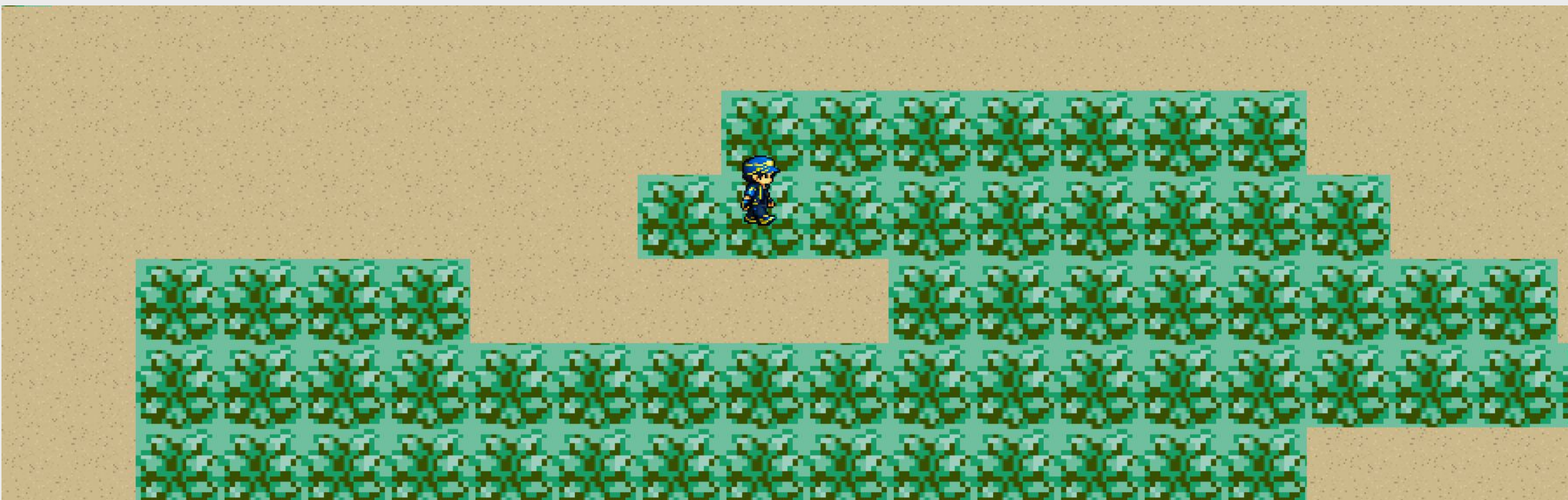
Codémon

Connor Allen, Isaac Hutchison, Lon Danna

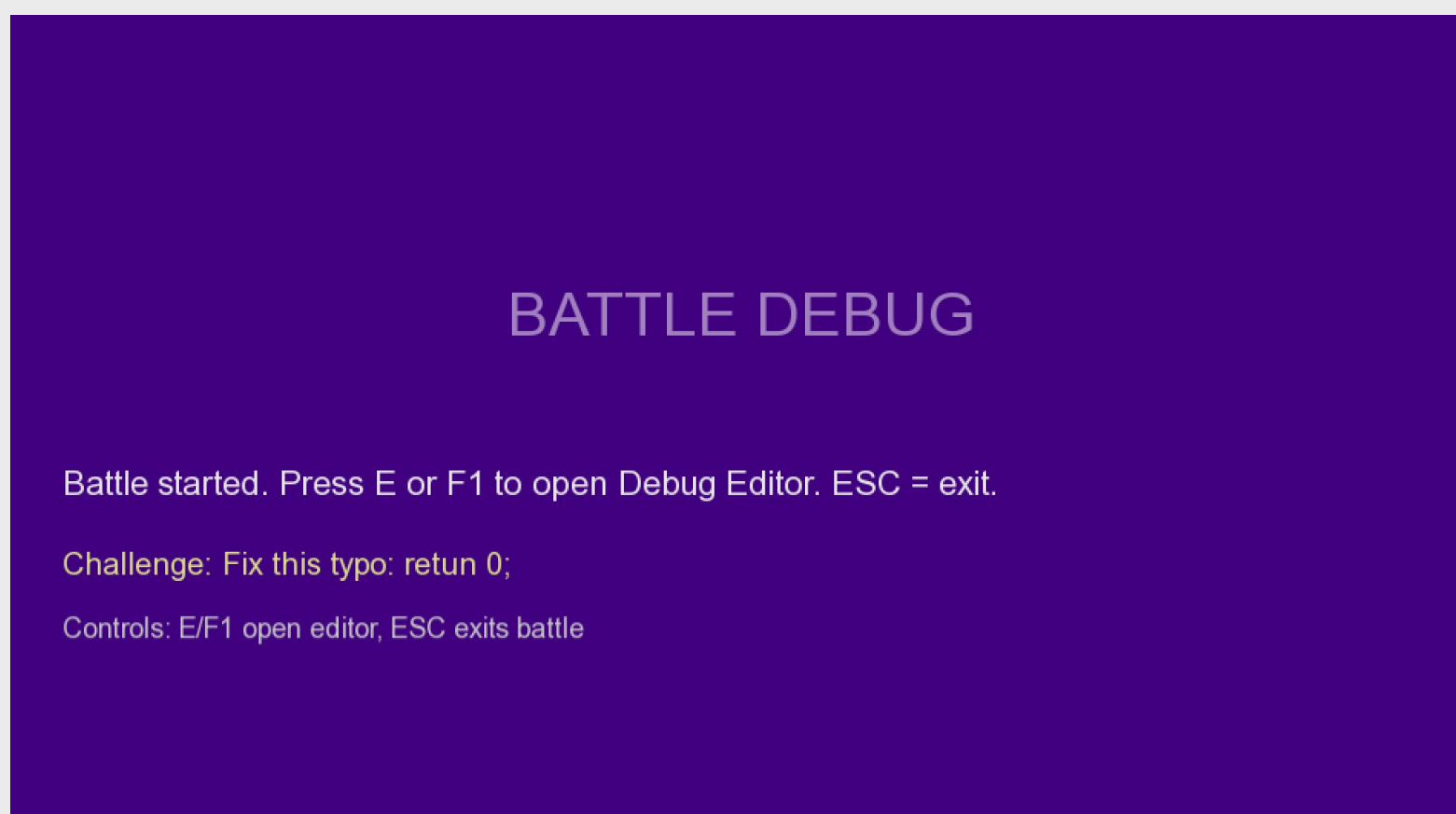
Overworld State



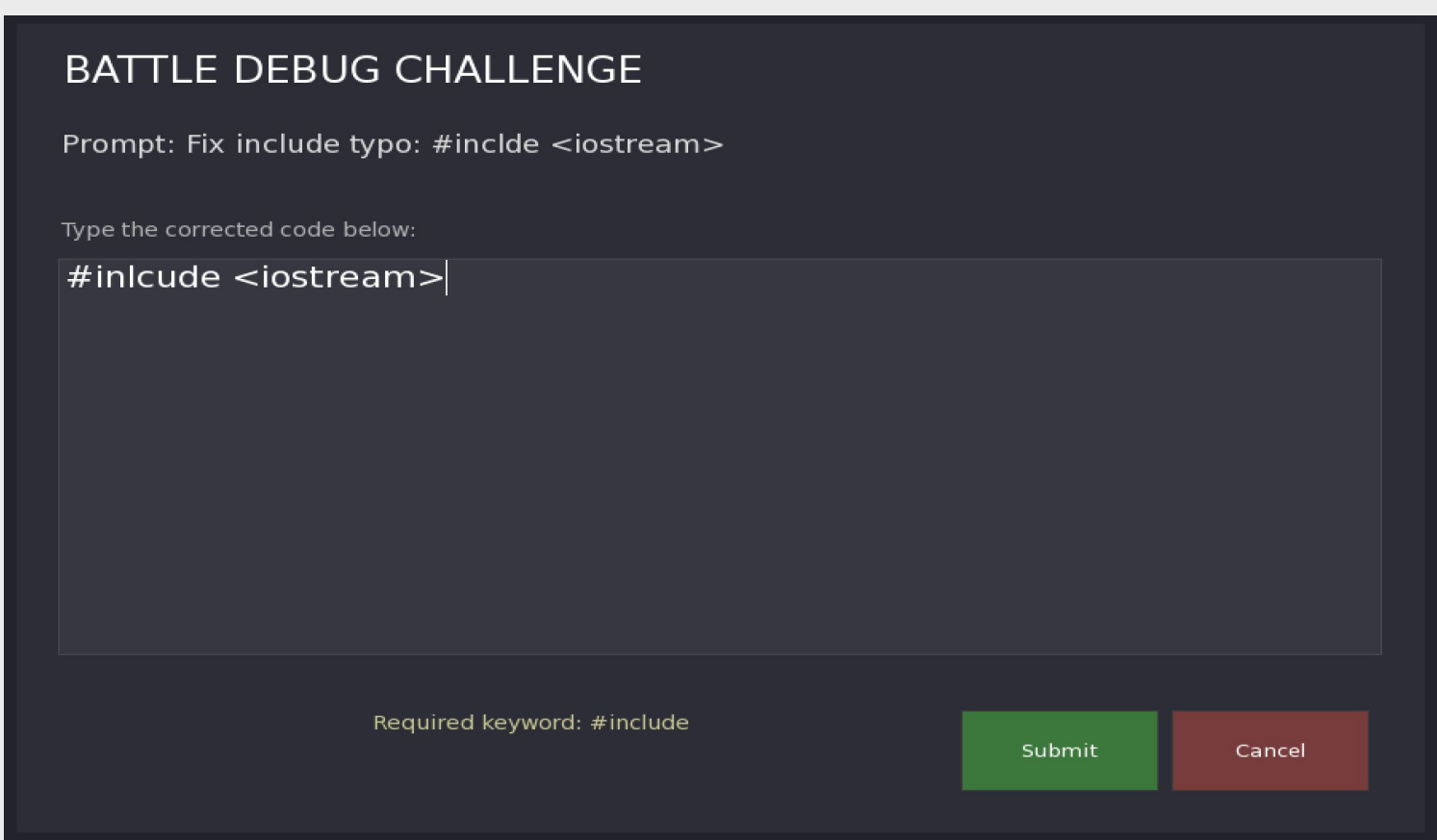
Battle State



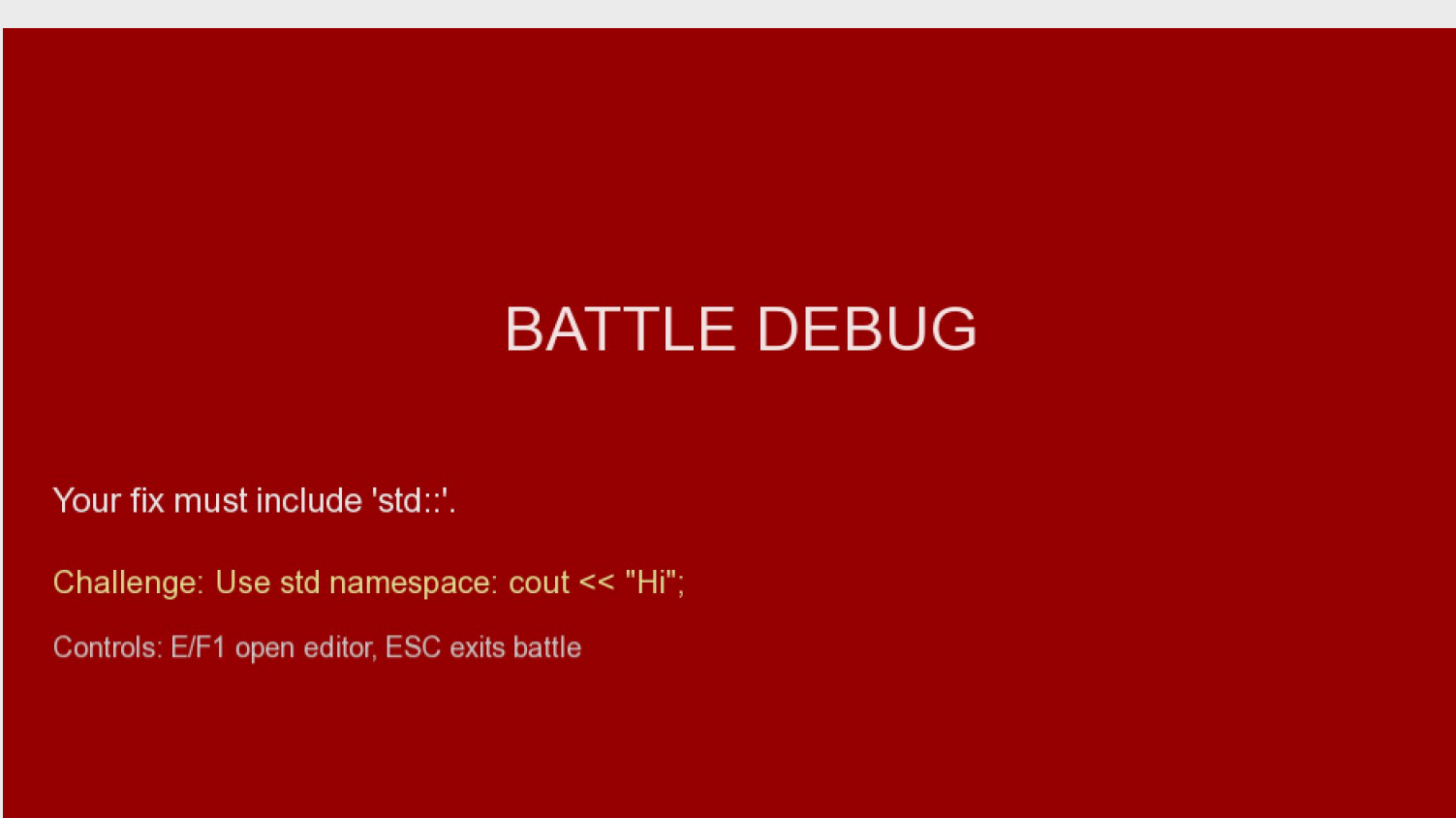
Battle State Engaged



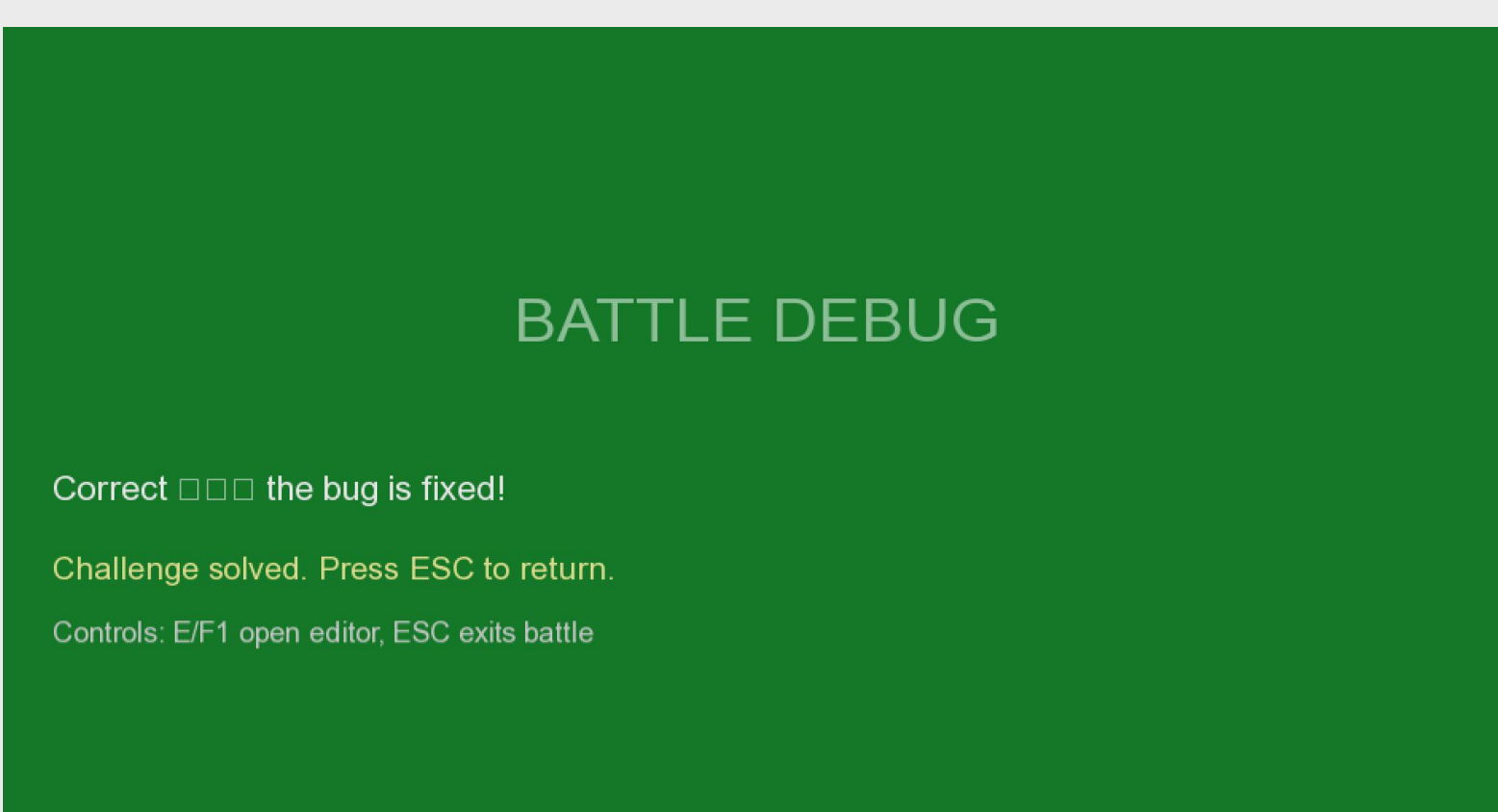
Debugging Challenge Interface



Incorrect Degubbing Sequence



Correct Debugging Sequence



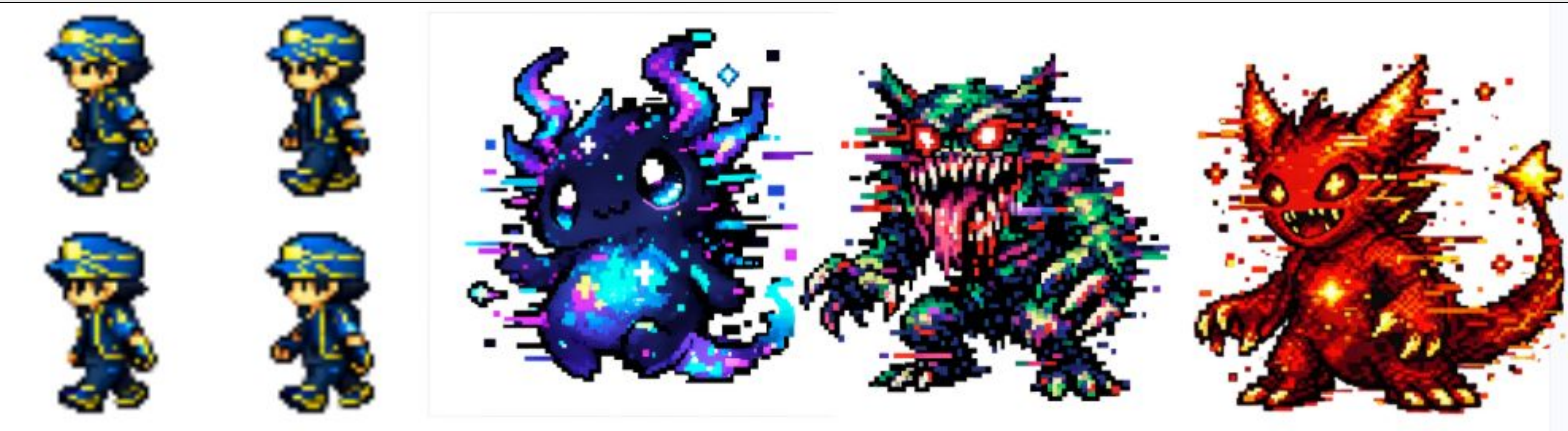
Abstract

Codémon: A Buggy World addresses the gap in CS education: the under-practice skill of debugging. While traditional tools code from scratch, students often struggle to diagnose existing errors. This project reframes debugging from a “failed state” to the core gameplay mechanics of a **C++ RPG**. Players progress by identifying and fixing logic, syntax, and resource errors embedded directly into the game’s creatures, battles, and environment.

Built with **C++** and **SFML**, Codémon uses a custom pattern-matching engine to validate player fixes in real time. By gamifying the diagnostics process—like correcting a boolean error to unlock a path—the system reduces **“debugging anxiety”** and build essential pattern recognition skills. This shift from code creation to code correction provide a realistic, **low-pressure** environment that better prepares students for the **problem-solving** demands of professional software development.

Codémon Features

- Debug-to-Win Battle System:** Battle various Codémon by debugging a snippet of code.
- Interactive Buggy Environment:** Parts of the world must be debugged to progress.
- Real-Time Syntax validation:** Players’ input is dynamically read and validated with pattern matching.
- Data-Driven Content (JSON):** Debugging challenges added and read dynamically with JSON.
- Integrated Terminal Overlay:** Using TGUI, a virtual terminal is displayed for the user to interact with.
- Cross-Platform Build System:** CMake automatically detects operating systems, ensuring compatibility with Windows, macOS and Linux.



GitHub QR Code



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